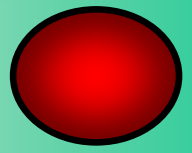
159.302 Tutorial



Propositional Logic

Tutorial Problem:

- **Use propositional resolution to solve a crime problem.**

Problem Definition

Solving a crime

- There are three suspects for a murder: Adams, Brown, and Clark.

Adams says "I didn't do it. The victim was an old acquaintance of Brown's. But Clark hated him."

Brown states "I didn't do it. I didn't know the guy. Besides I was out of town all week."

Clark says "I didn't do it. I saw both Adams and Brown in town around the victim that day; one of them must have done it."

- We know that exactly one of the suspects is guilty.
- Assume that the two innocent men are telling the truth, but that the guilty man might not be.

Knowledge Base

- The key to the solution is to make the suspect's statements conditional on their innocence. The following rules/axioms are sufficient.

If Adams was innocent, **then** Brown was the victim's friend and Clark did not like the victim, just as Adams said.

If Brown was innocent, **then** he wasn't in town and he doesn't know the victim.

If Clark was innocent, **then** Adams was with the victim and Brown was with the victim.

- **GENERAL KNOWLEDGE**

If Brown was with the victim, **then** Brown was in town.
If Brown was a friend of the victim, **then** Brown knew the victim.
If Clark liked the victim, **then** Clark knew the victim.

- **ASSERTION**

Now we assert that only one of A, B, and C is the guilty person:
A v B
A v C
B v C

One of the suspects is guilty

one of the suspects is guilty

EXPRESSION 1 $(\sim A \wedge B \wedge C) \vee (A \wedge \sim B \wedge C) \vee (A \wedge B \wedge \sim C)$

EXPRESSION 2 $(A \vee B) \wedge (A \vee C) \wedge (B \vee C)$

	A	B	C	EXPRESSION 1	EXPRESSION 2	
1	0	0	0	FALSE	FALSE	all are guilty
2	0	0	1	FALSE	FALSE	
3	0	1	0	FALSE	FALSE	
4	0	1	1	TRUE	TRUE	
5	1	0	0	FALSE	FALSE	
6	1	0	1	TRUE	TRUE	
7	1	1	0	TRUE	TRUE	
8	1	1	1	FALSE	TRUE	all are innocent

case 8 is assumed to be discarded in the crime case, as two are assumed to be innocent, not all of them.

Variables

A = Adams is innocent
B = Brown is innocent
C = Clark is innocent
X = Brown knew the victim
Y = Brown was in town
Z = Adams was out of town
W = Clark liked the victim
P = Brown was with the victim
Q = Adams was with the victim
F = Brown was a friend of the victim
S = Clark knew the victim

Axioms

A = Adams is innocent
B = Brown is innocent
C = Clark is innocent
X = Brown knew the victim
Y = Brown was in town
Z = Adams was out of town
W = Clark liked the victim
P = Brown was with the victim
Q = Adams was with the victim
F = Brown was a friend of the victim
S = Clark knew the victim

Translating the sentences, we have:

If Adams was innocent, **then** Brown was the victim's friend and Clark did not like the victim, just as Adams said.	$A \rightarrow (F \wedge \sim W)$
If Brown was innocent, **then** he wasn't in town and he doesn't know the victim.	$B \rightarrow (\sim Y \wedge \sim X)$
If Clark was innocent, **then** Adams was with the victim and Brown was with the victim.	$C \rightarrow (Q \wedge P)$
If Brown was with the victim, **then** Brown was in town.	$P \rightarrow Y$
If Brown was a friend of the victim, **then** Brown knew the victim.	$F \rightarrow X$
If Clark liked the victim, **then** Clark knew the victim.	$W \rightarrow S$
A v B A v C B v C	$A \vee B$ $A \vee C$ $B \vee C$

Next, we shall expand the sentences…

Axioms

CNF of the sentences:

	Eliminating arrows	Distributing OR over AND
A→(F ∧ ~W)	~A ∨ (F ∧ ~W)	(~A ∨ F) ∧ (~A ∨ ~W)
B→(~Y ∧ ~X)	~B ∨ (~Y ∧ ~X)	(~B ∨ ~Y) ∧ (~B ∨ ~X)
C→(Q ∧ P)	~C ∨ (Q ∧ P)	(~C ∨ Q) ∧ (~C ∨ P)
P→Y	~P ∨ Y	~P ∨ Y
F→X	~F ∨ X	~F ∨ X
W→S	~W ∨ S	~W ∨ S
A ∨ B		A ∨ B
A ∨ C		A ∨ C
B ∨ C		B ∨ C

Axioms

Splitting the sentences, we have:

(~A ∨ F) ∧ (~A ∨ ~W)	(~A ∨ F)
	(~A ∨ ~W)
(~B ∨ ~Y) ∧ (~B ∨ ~X)	(~B ∨ ~Y)
	(~B ∨ ~X)
(~C ∨ Q) ∧ (~C ∨ P)	(~C ∨ Q)
	(~C ∨ P)
~P ∨ Y	~P ∨ Y
~F ∨ X	~F ∨ X
~W ∨ S	~W ∨ S
A ∨ B	A ∨ B
A ∨ C	A ∨ C
B ∨ C	B ∨ C

Axioms

Pick one of the suspects, then attempt to prove that he is not innocent (e.g. Brown)

- (~A ∨ F)
- (~A ∨ ~W)
- (~B ∨ ~Y)
- (~B ∨ ~X)
- (~C ∨ Q)
- (~C ∨ P)
- ~P ∨ Y
- ~F ∨ X
- ~W ∨ S
- A ∨ B
- A ∨ C
- B ∨ C
- **B**

Brown is not innocent ≡ ~B

First, we negate the conclusion, then we add it to our axioms…

A = Adams is innocent
B = Brown is innocent
C = Clark is innocent
X = Brown knew the victim
Y = Brown was in town
Z = Adams was out of town
W = Clark liked the victim
P = Brown was with the victim
Q = Adams was with the victim
F = Brown was a friend of the victim
S = Clark knew the victim

Axioms

Pick one of the suspects, then attempt to prove that he is not innocent (e.g. Brown)

1	(~A v F)
2	(~A v ~W)
3	(~B v ~Y)
4	(~B v ~X)
5	(~C v Q)
6	(~C v P)
7	~P v Y
8	~F v X
9	~W v S
10	A v B
11	A v C
12	B v C
13	B

Proof Steps:

14	~X	13,4

Axioms

Pick one of the suspects, then attempt to prove that he is not innocent (e.g. Brown)

1	(~A v F)
2	(~A v ~W)
3	(~B v ~Y)
4	(~B v ~X)
5	(~C v Q)
6	(~C v P)
7	~P v Y
8	~F v X
9	~W v S
10	A v B
11	A v C
12	B v C
13	B

Proof Steps:

14	~X	13,4
15	~F	14,8
16	~A	15,1
17	C	16,11
18	P	17,6
19	Y	18,7
20	~B	19,3
21	■	20,13

Therefore, Brown is guilty!

You can try to prove that Adam or Clark is guilty, but the proof system will not lead to a contradiction.